

UNIT -II

FORM AND ORIENTATION TOLERANCE

Introduction to geometric dimensioning and tolerancing:

What is GD&T?

GD&T, short for Geometric Dimensioning and Tolerancing, is a system for defining and communicating design intent and engineering tolerances that helps engineers and manufacturers optimally control variations in manufacturing processes.

Limitations of Tolerancing Before GD&T

- Before GD&T, manufacturing features were specified by X-Y areas. For example, when drilling a mounting hole, the hole had to be within a specified X-Y area.
- An accurate tolerancing specification, however, would define the position of the hole in relation to the intended position, the accepted area being a circle. X-Y tolerancing leaves a zone in which inspection would have produced a false negative because while the hole is not within the X-Y square, it would fall within the circumscribed circle.
- Stanley Parker, an engineer who was developing naval weapons during World War II, noticed this failure in 1940. Driven by the need for cost-effective manufacturing and meeting deadlines, he worked out a new system through several publications. Once proven as a better operational method, the new system became a military standard in the 1950s.
- Currently, the GD&T standard is defined by the American Society of Mechanical Engineers ([ASME Y14.5-2018](#)) for the USA and [ISO 1101-2017](#) for the rest of the world. It concerns mostly the overall geometry of the product, while other standards describe specific features such as surface roughness, texture, and screw threads.

Why Implement GD&T

GD&T is crucial for functional assemblies, multi-part products, or parts with complex functionality. With functional assemblies, multi-part products, or parts with complex functionality, it is crucial that all components work well together. All relevant fits and features need to be specified in a way that impacts the manufacturing process and its related investments the least, while still guaranteeing functionality. Tightening tolerances by a factor two can raise the costs twofold or even more, due to higher reject rates and tooling changes. GD&T is the system that allows developers and inspectors to optimize functionality without increasing cost.

The most important benefit of GD&T is that the system describes the design intent rather than the resulting geometry itself. Like a vector or formula, it is not the actual object but a representation of it.

- For example, a feature standing at 90 degrees to a base surface can be toleranced on its perpendicularity to that surface. This will define two planes spaced apart, that the center plane of the feature must fall within. Or, when drilling a hole, it makes the most sense to tolerance it in terms of alignment to other features.
- Describing product geometry related to its intended functionality and manufacturing approach is ultimately simpler than having to describe everything in linear dimensions. It also provides a communication tool with manufacturing vendors, customers, as well as quality inspectors.
- When performed well, GD&T even allows statistical process control (SPC), reducing product reject rates, assembly failures, and the effort needed for quality control, saving organizations substantial resources. As a result, multiple departments are able to work more in parallel because they have a shared vision and language for what they want to achieve.

How GD&T Works

- Engineering drawings need to show the dimensions for all features of a part. Next to the dimensions, a tolerance value needs to be specified with the minimum and maximum acceptable limit. The tolerance is the difference between the minimum and maximum limit. For example, if we have a table that we would accept with a height between 750 mm and 780 mm, the tolerance would be 30 mm.
- However, the tolerance for the table implies that we would accept a table that is 750 mm high on one side and 780 mm on the other, or has a waved surface with 30 mm

variation. So to appropriately tolerance the product, we need a symbol communicating the design intent of a flat top surface. Therefore we have to include an additional flatness tolerance in addition to the overall height tolerance.

- Parts with unpredictable variations and complex shapes require GD&T practices beyond simple plus-minus tolerancing.
- Similarly, a cylinder with a toleranced diameter will not necessarily fit into its hole if the cylinder gets slightly bent during the manufacturing process. Therefore it also needs a straightness control, which would be difficult to communicate with traditional plus-minus tolerancing. Or a tube that has to seamlessly match a complex surface that it's welded to requires a surface profile control.
- GD&T establishes a library of symbols to convey such design intents, which we discuss in the following section.
- Dynamic assemblies such as this prosthetic hand require precise tolerancing.
- The art of tolerancing means to specify just the right variations for all specific design features in order to maximize product approval rate within the limits of the manufacturing processes and depending on the part's visual and functional purpose.
- In the metric system, there are International Tolerance (IT) grades that can also be used to specify tolerances by means of symbols. The symbol 40H11, for example, means a 40 mm diameter hole with a loose running fit.
- The manufacturer then only needs to look up the basis table for hole features to derive the exact tolerance value.
- Besides individual tolerances, engineers must take into account system-level effects. For example, when a part comes out with all dimensions at their maximum allowed value, does it still meet overall requirements such as product weight and wall thicknesses? This is called the **Maximum Material Condition** (MMC), while its counterpart is the **Least Material Condition** (LMC).
- Tolerances also stack up. If we create a chain link where each hole has a 0.1 mm plus tolerance and each shaft a 0.1 mm negative tolerance, that means we will still accept a 20 mm length difference at 100 links. When installing repeated elements such as a perforated hole pattern, first position the pattern and then specify interrelated distances rather than referencing elements to a fixed edge or plane of the part.

- The standards do not only pertain to designers and engineers but also to quality inspectors by informing them how to measure the dimensions and tolerances. Using specific tools such as digital micrometers and calipers, height gauges, surface plates, dial indicators, and a coordinate measuring machine (CMM) are important to tolerancing practice.
- When measuring and defining a part, the geometry exists in a conceptual space called the Datum Reference Frame (DRF). This is comparable to the coordinate system at the origin of a space in 3D modeling programs. A datum is a point, line or plane that exists in the DRF and is used as a starting place for measuring. Make sure to define the datum features relevant to the functionality of your part. Unless you are mating features of one part to those of others in an assembly, you can often use a single datum. Always make sure that the primary datum has a reliable location to derive other measurements from, for example, where the final part will have little unpredictable variation.

GD&T Tolerancing Guidelines

An engineering drawing has to accurately convey the product without adding unnecessary complexity or restrictions. The following guidelines are helpful to consider:

- Clarity of a drawing is the most important, even more so than its accuracy and completeness. To improve clarity, draw dimensions and tolerances outside of the part's boundaries and applied to visible lines in true profiles, employ a unidirectional reading direction, convey the function of the part, group and/or stagger dimensions, and make use of white space.
- Always design for the loosest feasible tolerance to keep costs down.
- Use a general tolerance defined at the bottom of the drawing for all dimensions of the part. Specific tighter or looser tolerances indicated in the drawing will then supersede the general tolerance.
- Tolerance functional features and their interrelations first, then move on to the rest of the part.
- Whenever possible, leave GD&T work to the manufacturing experts and do not describe manufacturing processes in the engineering drawing.
- Do not specify a 90-degree angle since it is assumed.
- Dimensions and tolerances are valid at 20 °C / 101.3 kPa unless stated otherwise.

GD&T Symbols

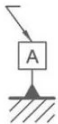
GD&T is feature-based, with each feature specified by different controls. GD&T symbols fall into five groups:

- **Form controls** specify the shape of features, including:
 - *Straightness* is divided into line element straightness and axis straightness.
 - *Flatness* means straightness in multiple dimensions, measured between the highest and lowest points on a surface.
 - *Circularity* or roundness can be described as straightness bent into a circle.
 - *Cylindricity* is basically flatness bent into a barrel. It includes straightness, roundness, and taper, which makes it expensive to inspect.
- **Profile controls** describe the three-dimensional tolerance zone around a surface:
 - *Line Profile* compares a two-dimensional cross-section to an ideal shape. The tolerance zone is defined by two offset curves unless otherwise specified.
 - *Surface Profile* creates through two offset surfaces between which the feature surface must fall. This is a complex control typically measured with a CMM.
- **Orientation controls** concern dimensions that vary at angles, including:
 - *Angularity* is flatness at an angle to a datum and is also determined through two reference planes spaced the tolerance value apart.
 - *Perpendicularity* means flatness at 90 degrees to a datum. It specifies two perfect planes the feature plane must lie in between.
 - *Parallelism* means straightness at a distance. Parallelism for axes can be defined by defining a cylindrical tolerance zone by placing a diameter symbol in front of the tolerance value.
- **Location controls** define feature locations using linear dimensions:
 - *Position* is the location of features relative to one another or to datums and is the most used control.
 - *Concentricity* compares the location of a feature axis to the datum axis.
 - *Symmetry* ensures that non-cylindrical parts are similar across a datum plane. This is a complex control typically measured with a CMM.
- **Runout controls** define the amount by which a particular feature can vary with respect to the datums:

- *Circular Runout* is used when there is a need to account for many different errors, such as ball-bearing mounted parts. During inspection, the part is rotated on a spindle to measure the variation or ‘wobble’ around the rotational axis.
- *Total Runout* is measured on multiple points of a surface, not just describing the runout of a circular feature but of an entire surface. This controls straightness, profile, angularity, and other variations.

GD&T Symbols Overview :

APPLICATION	TYPE	CHARACTERISTIC	SYMBOL
Individual Features	Form	Straightness	
		Flatness	
		Circularity	
		Cylindricity	
Individual or Related Features	Profile	Line Profile	
		Surface Profile	
Related Features	Orientation	Angularity	
		Perpendicularity	
		Parallelism	
	Location	Position	
		Concentricity	
		Symmetry	
	Runout	Circular Runout	
		Total Runout	



Datum



 0.25 



Feature Control Frame

Both ANSI and ISO standards use these common symbols for tolerancing controls.

Feature Control Frame (FCF)

The Feature Control Frame is the notation to add controls to the drawing. The leftmost compartment contains the geometric characteristic. In the example above, it is a location control but it can contain any of the control symbols. The first symbol in the second compartment indicates the shape of the tolerance zone. In this example, it is a diameter as opposed to a linear dimension. The number indicates the allowed tolerance.

Next to the tolerance box, there are separate boxes for each datum feature that the control refers to. Here, the location will be measured related to datum B and C. Next to the tolerance or a datum feature is an optional encircled letter, the feature modifier.

The following possibilities can occur:

- **M** means that the tolerance applies in the Maximum Material Condition (MMC)
- **L** means that the tolerance applies in the Least Material Condition (LMC)
- **U** indicates an unequal bilateral tolerance, i.e. for a 1 mm tolerance it may specify it as minus 0.20 and plus 0.80.
- **P** means that the tolerance is measured in a Projected Tolerance Zone at a specified distance from the datum.
- No symbol installs the tolerance regardless of feature size (RFS)

For this example, if the part is not in MMC, a bonus tolerance can be added proportionally to the deviation from MMC. So if a part is at 90% MMC, the tolerance will also loosen by 10%.

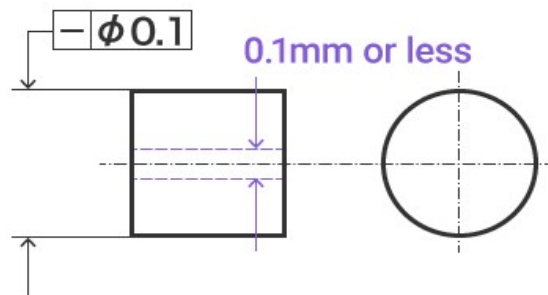
Types of form tolerances:

Straightness



The straightness requirement specifies how perfectly straight a target should be. It is applied to lines and not planes, and represents a curve in the center line or generatrix. Therefore, straightness is used to indicate the warpage tolerance of long objects.

Sample indication:



Explanation of drawings:

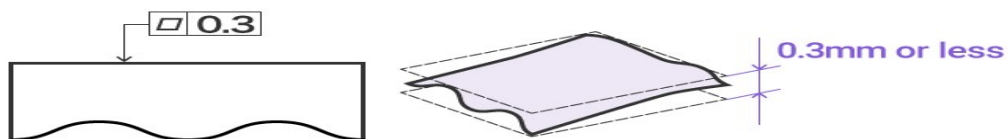
If a tolerance frame is connected to the size indicating the diameter of a cylinder, the axis of that cylinder must be within a cylinder with a diameter of 0.1 mm.

Flatness



The flatness requirement specifies the evenness of a surface, or how accurately flat a target plane should be. The most protruding part and the most concaved part must be at a specific distance between two planes that are separated vertically.

Sample indication:



Explanation of drawings:

This surface must be between two parallel planes separated by only 0.3 mm.

Roundness



The roundness requirement specifies how perfectly circular a target—the circular cross-section of a shaft, bore, or cone—should be.

Sample indication:



Explanation of drawings:

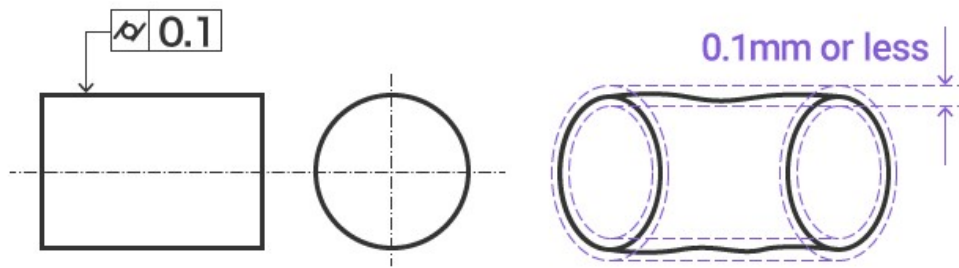
The outer circumference of any cross-section of a shaft cut perpendicularly should be between two concentric circles just 0.1 mm apart on the same plane.

Cylindricity



The cylindricity requirement specifies how accurately circular and straight a target cylinder is. The value represents any distortion in a cylinder.

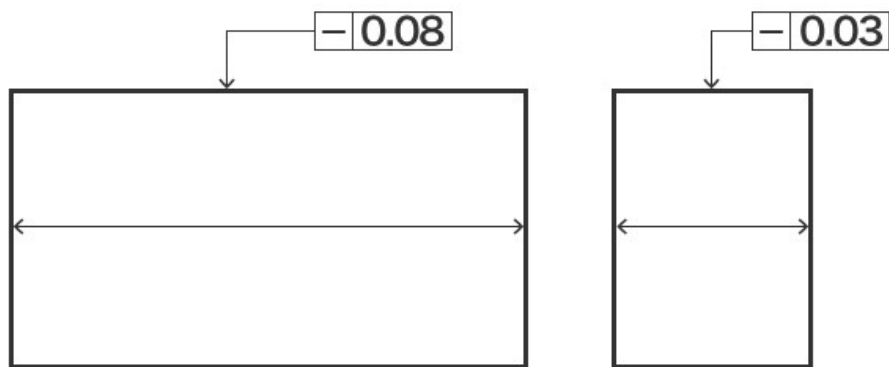
Sample indication:



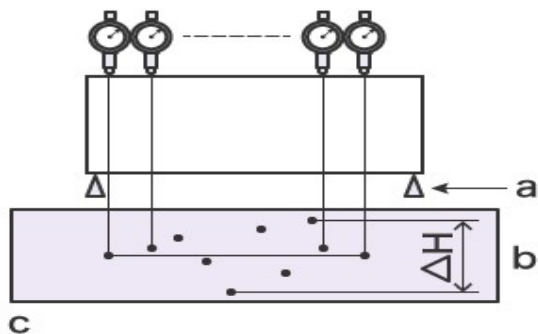
Explanation of drawings:

The target plane must be between two coaxial cylinders just 0.1 mm apart.

Measurement of straightness:



Using a Height Gauge



a) Micro jack

b) ΔH = straightness

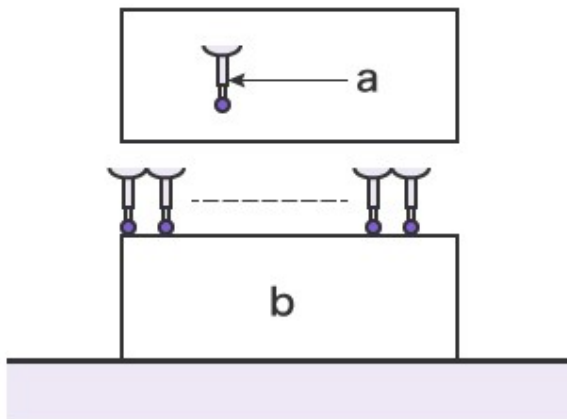
c) Height (H_n) graph

- Secure the target so that the height is evenly matched on the left and right, using small jacks in order to prevent the target from tilting. Move the target or the height gauge straight to measure the straightness. The difference between the maximum and minimum values (ΔH) is the straightness.

DISADVANTAGES

In general, height gauges have lower precision than coordinate measuring machines. In addition, measured values can change with the force used to place the measuring part of the height gauge on the target, causing the measurement results to become unstable. With targets that cannot be placed level, the height gauge cannot be moved, which makes measurement difficult.

Using a Coordinate Measuring Machine (CMM)



a) Stylus

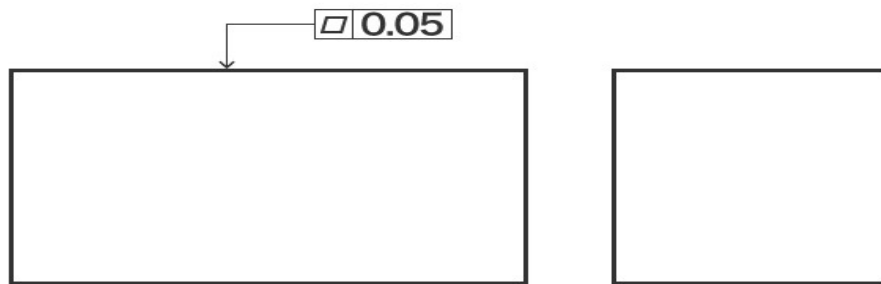
b) Target

- A coordinate measuring machine can measure straightness by the operator simply putting the stylus lightly on the target. Thanks to this feature, there is almost no error

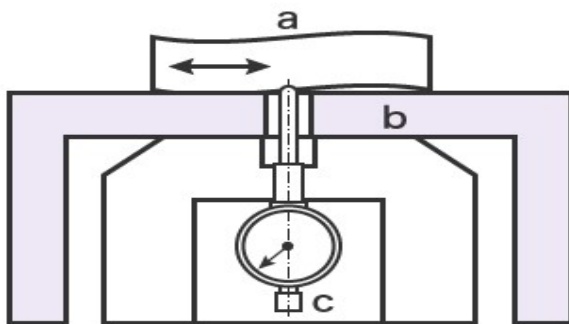
caused by measurement pressure, and stable measurement results can be obtained. Additionally, the stylus can be placed on the target from different angles, which enables accurate measurement of targets that cannot be secured level, i.e., targets that cannot be measured using a height gauge.

Measurement of flatness:

- When measuring [flatness](#), you are checking for unevenness in the surface, to see how precisely flat a surface is the most protruding part and the most concaved part must be at a specific distance between two planes that are separated vertically.



Using a Dial Gauge



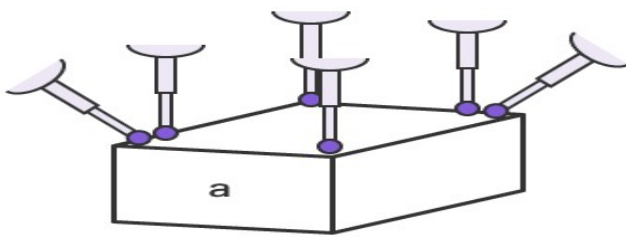
- a) Target
- b) Table
- c) Dial gauge

- Place the target on the precision plane table and secure it in place. Set the dial gauge so that its measuring part comes into contact with the measurement surface. Move the target so that the measurement surface is evenly measured, and read the dial gauge values. The largest deviation value is the flatness.

DISADVANTAGES

The measured values may change depending on any changes in the measurement points, since the measurement points change depending on how you move the target. Therefore, it is difficult to obtain stable measured values.

Using a Coordinate Measuring Machine (CMM)



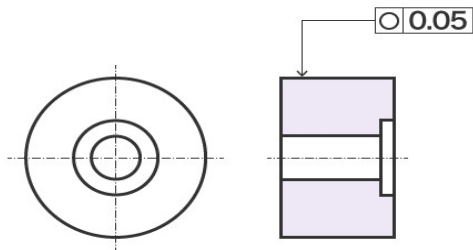
a) Target

- Put the stylus on four or more points for point measurement of flatness. Increasing the number of measurement points enables precise, stable measurement even for wide measurement ranges.

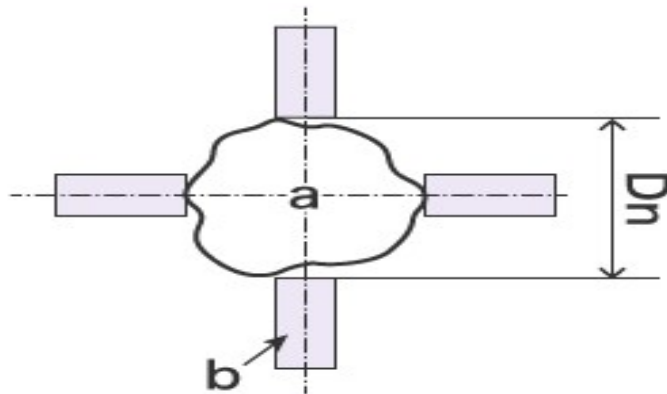
Measurement of Roundness:

- When measuring [roundness](#), you are measuring the roundness of a circular cross-section of a shaft, bore, or circular cone, to see how accurate a circle it is.

Sample Drawings



Using a Micrometer



a) Target

b) Micrometer measurement terminal

- Two-point measurement is performed on the outer form by dividing it into four to eight sections. The roundness is the value obtained by dividing the difference between the maximum and minimum values by 2. A micrometer is all that is needed for measurement; you can take measurements easily, anywhere.

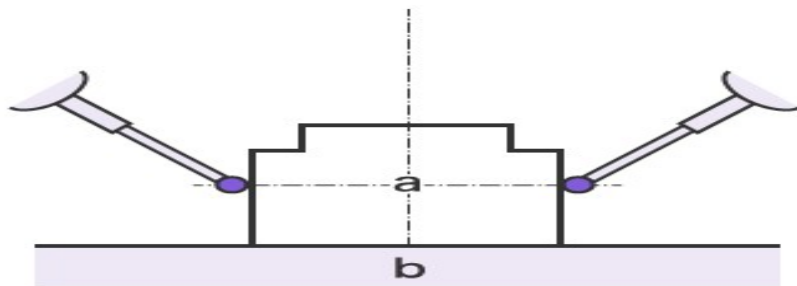
- $\text{Roundness} = (D_{\text{max}} - D_{\text{min}}) / 2$

- Dmax: Maximum value
- Dmin: Minimum value

DISADVANTAGES

There is a limit to the accuracy of the measurement, as the value is an average of two-point measurements. Additionally, an error may occur due to the differences in how the micrometer measurement terminal is placed on the target. To analyze the obtained data, you also need to enter it into a computer.

Using a Coordinate Measuring Machine (CMM)



- a) Target
- b) Surface plate

- Put the stylus on four or more points for point measurement of roundness. The measured value is calculated by [least-squares method](#), where the sum of the deviations from all measurement points is the smallest.

Orientation Tolerance:

- Orientation tolerance determines the orientation for the form in relation to a reference. A [datum](#) is always necessary to indicate orientation tolerance; as such, it is a geometric tolerance for features related to datums.

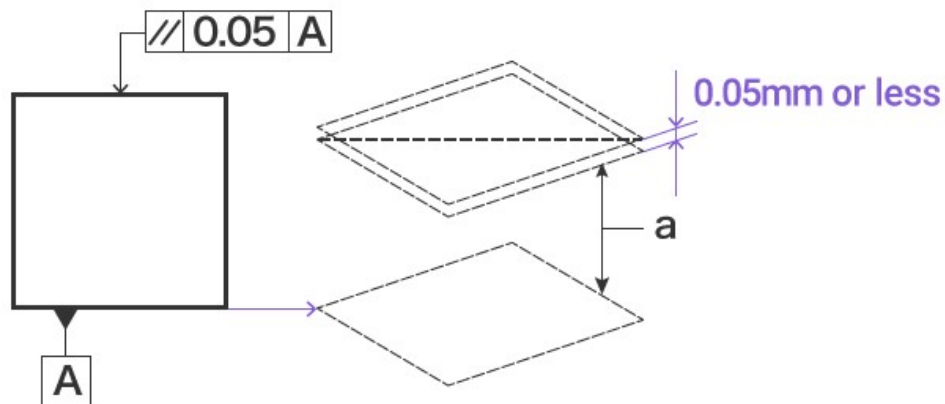
Types of orientation Tolerances:

Parallelism



While it appears to be similar to [flatness](#), parallelism involves a datum (reference plane or line). The parallelism requirement specifies that two lines or planes are parallel.

Sample indication:



a) Parallel

Explanation of drawings:

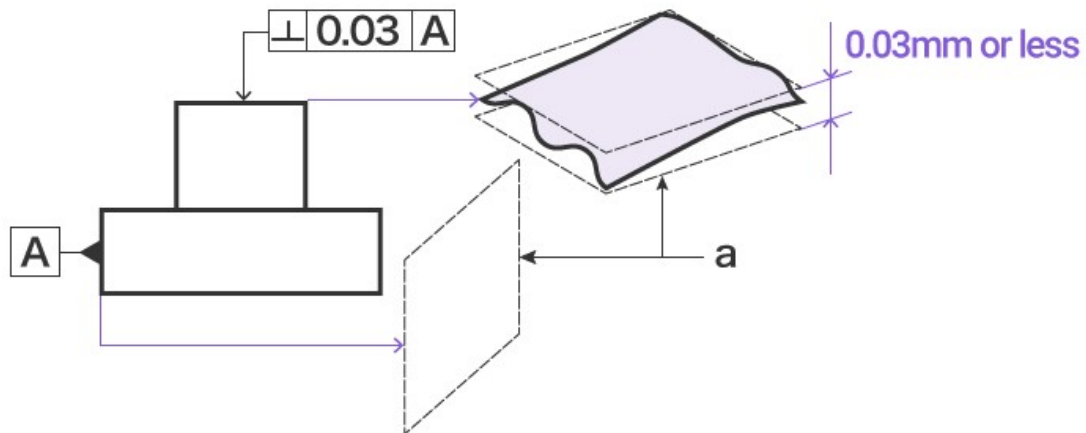
The plane indicated by the indication arrow must be parallel to datum plane A and be between two planes only 0.05 mm apart in the direction of the indication arrows.

Perpendicularity



The perpendicularity requirement specifies the accuracy of how perpendicular a target is to the datum (reference plane or line). Instead of degrees, millimeters are used as the unit for the value of perpendicularity indication.

Sample indication:



a) Right angle

Explanation of drawings:

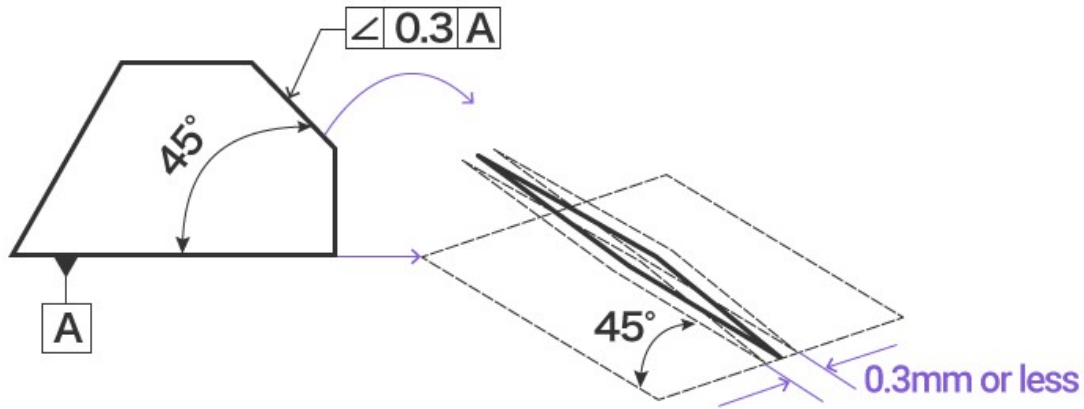
The plane indicated by the indication arrow must be within two parallel planes perpendicular to datum plane A and with a diameter of 0.03 mm.

Angularity



The angularity requirement specifies how accurate an angle is to the datum (reference plane or line) when the specified line or plane is not 90 degrees. Instead of degrees, the millimeter is used as the unit for the value of angularity indication.

Sample indication:



Explanation of drawings:

The plane indicated by the indication arrow must be theoretically exactly angled by 45 degrees to the datum plane A and between two parallel planes only 0.3 mm apart in the direction of the indication arrows.

verification of orientation tolerances in GD&T :

In Geometric Dimensioning and Tolerancing (GD&T), orientation tolerances play a crucial role in defining the acceptable variations in the orientation of features on a part. These tolerances ensure that the part functions as intended and can be manufactured within specified limits. Verifying orientation tolerances in GD&T typically involves the use of various measurement and inspection methods. Here are some common techniques:

1. Coordinate Measurement Machines (CMM):

- CMMs are sophisticated devices that use a probe to measure the coordinates of points on a part's surface.
- Software on the CMM can then analyze these measurements to determine the orientation of features relative to the specified tolerances.

2. Profile Projectors:

- Profile projectors use optical magnification to project an enlarged image of a part onto a screen.
- This allows inspectors to visually compare the part's features to a template or overlay with the specified orientation tolerances.

3. Height Gauges:

- Height gauges can be used to measure the distance between a reference surface and the feature of interest.
- By comparing the measured distance to the specified tolerance, inspectors can verify the orientation of the feature.

4. **Gauge Blocks and V-Blocks:**

- Gauge blocks and V-blocks can be employed to set up parts in a way that facilitates the measurement of orientation.
- These blocks provide known reference surfaces against which the part can be aligned for measurement.

5. **Surface Plates and Dial Indicators:**

- Surface plates provide a flat and stable surface for placing the part.
- Dial indicators can then be used to measure deviations from the desired orientation by assessing the part's position relative to the specified tolerances.

6. **3D Scanning:**

- 3D scanning technologies, such as laser scanners or structured light scanners, can capture the entire surface geometry of a part.
- The scanned data can be compared to a CAD model, and software can analyze the orientation of features.

7. **Optical Comparators:**

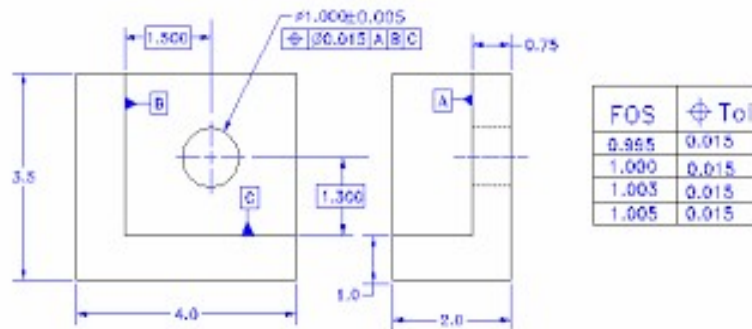
- Optical comparators use a light source and lenses to project a magnified silhouette of a part onto a screen.
- This allows for visual inspection of features against specified tolerances.

8. **Interferometry:**

- Interferometry involves the use of interference patterns to measure distances and angles with high precision.
- Interferometers can be used to assess the orientation of surfaces and features.

It's important to note that the choice of verification method depends on factors such as the specific GD&T requirements, the complexity of the part, and the available equipment. Manufacturers often use a combination of these methods to ensure accurate and reliable verification of orientation tolerances in GD&T. Additionally, adherence to industry standards and guidelines, such as those provided by the ASME Y14.5 standard for GD&T, is essential for proper orientation tolerance verification.

RFS Concept for form tolerances:



In Geometric Dimensioning and Tolerancing (GD&T), the concept of RFS stands for "Regardless of Feature Size." When the symbol "RFS" is specified in the feature control frame, it indicates that the tolerance values apply without regard to the size of the feature being controlled. This is primarily associated with form tolerances, and it ensures that the specified tolerance is independent of the actual size of the feature.

For example, if you have a flatness tolerance with the RFS modifier, it means that the flatness tolerance applies uniformly across the entire surface, regardless of the size of the part. The goal is to ensure that the specified form tolerance does not increase or decrease with the size of the feature.

Here's how the RFS concept works for some common form tolerances:

1. Flatness (F):

- If a flatness tolerance is specified as 0.1 RFS, it means that the flatness requirement applies uniformly across the entire surface of the feature, regardless of its size.

2. Straightness (S):

- Similarly, for straightness with the RFS modifier, the specified straightness tolerance applies uniformly along the entire length of the feature, regardless of its size.

3. **Circularity (C):**

- In circularity, the RFS modifier means that the tolerance for roundness applies regardless of the diameter of the circular feature.

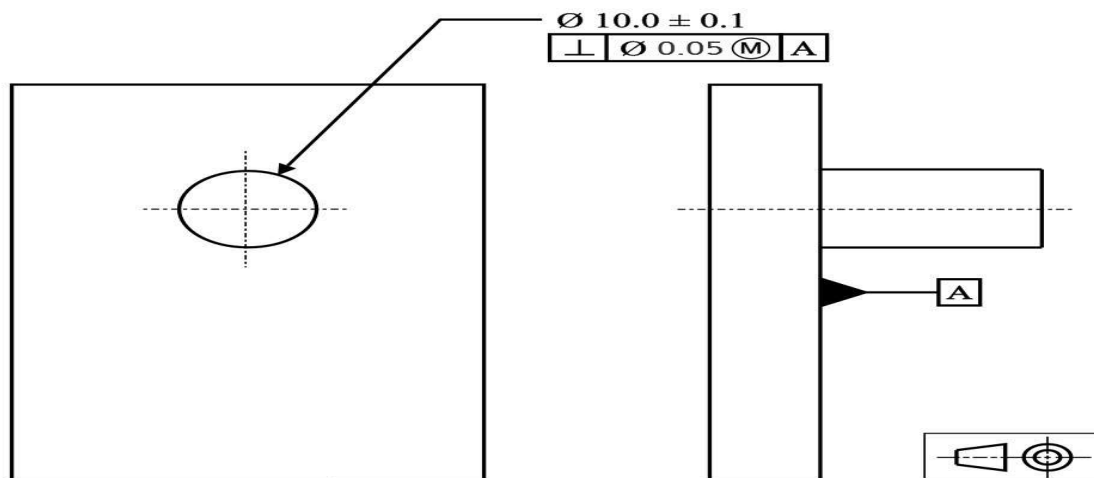
4. **Cylindricity (Cyl):**

- Cylindricity with the RFS modifier ensures that the specified tolerance for the overall form of the cylindrical feature applies without considering its size.

The RFS concept is important because it allows for consistent interpretation of form tolerances across different-sized features. It simplifies the drawing and inspection process by eliminating the need to adjust tolerances based on the size of the part.

Keep in mind that the RFS concept is just one approach to specifying form tolerances in GD&T. In some cases, other modifiers like "M" (Maximum Material Condition) or "L" (Least Material Condition) may be used to modify the tolerance based on the actual size of the feature. The choice of modifier depends on the design intent and the functional requirements of the part.

MMC Concept for Form Tolerances:



In Geometric Dimensioning and Tolerancing (GD&T), MMC stands for Maximum Material Condition. When MMC is applied to form tolerances in a feature control frame, it means that the specified tolerance applies when the feature contains the maximum amount of material permissible within the specified limits. In other words, MMC is used to control the form of a feature when it is at its largest size (maximum material condition).

For form tolerances, such as flatness, straightness, circularity, or cylindricity, specifying MMC in the feature control frame implies that the tolerance is only applicable when the feature contains the maximum amount of material. This concept is often used to ensure that parts with varying sizes meet specific form requirements.

Here's how MMC works for some common form tolerances:

1. Flatness (F):

- If a flatness tolerance is specified with MMC, it means that the flatness requirement applies only when the feature is at its maximum material condition. In other words, the surface must be as flat as possible when the part is at its largest size.

2. Straightness (S):

- Similarly, for straightness with MMC, the specified straightness tolerance applies when the feature is at its maximum material condition, ensuring that the feature is as straight as possible under these conditions.

3. Circularity (C):

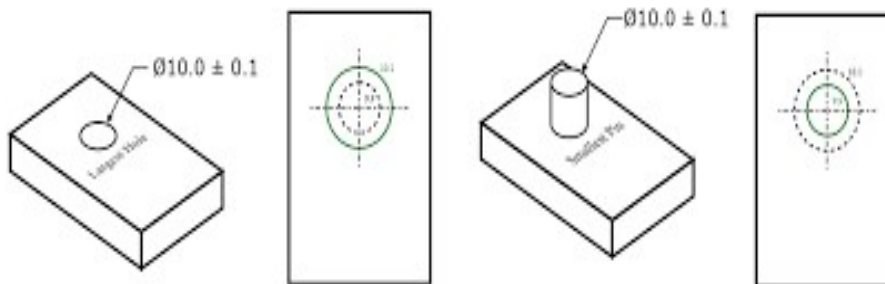
- Circular features with MMC circularity tolerances must conform to the specified form requirements when the feature is at its maximum material condition, regardless of its actual size.

4. **Cylindricity (Cyl):**

- Cylindricity with MMC ensures that the specified tolerance for the overall form of the cylindrical feature applies when the feature is at its maximum material condition.

The MMC concept is often used when the function of the part requires a certain level of form control, especially in situations where variations in size might affect the part's performance. It is important to note that MMC is just one modifier used in GD&T, and the choice of modifier depends on the design intent and functional requirements of the part.

LMC Concept for Form tolerances:



In Geometric Dimensioning and Tolerancing (GD&T), LMC stands for Least Material Condition. When LMC is applied to form tolerances in a feature control frame, it means that the specified tolerance applies when the feature contains the least amount of material permissible within the specified limits. In other words, LMC is used to control the form of a feature when it is at its smallest size (least material condition).

For form tolerances, such as flatness, straightness, circularity, or cylindricity, specifying LMC in the feature control frame implies that the tolerance is only applicable when the feature contains the least amount of material. This concept is often used to ensure that parts with varying sizes meet specific form requirements.

Here's how LMC works for some common form tolerances:

1. **Flatness (F):**

- If a flatness tolerance is specified with LMC, it means that the flatness requirement applies only when the feature is at its least material condition. In other words, the surface must be as flat as possible when the part is at its smallest size.

2. Straightness (S):

- Similarly, for straightness with LMC, the specified straightness tolerance applies when the feature is at its least material condition, ensuring that the feature is as straight as possible under these conditions.

3. Circularity (C):

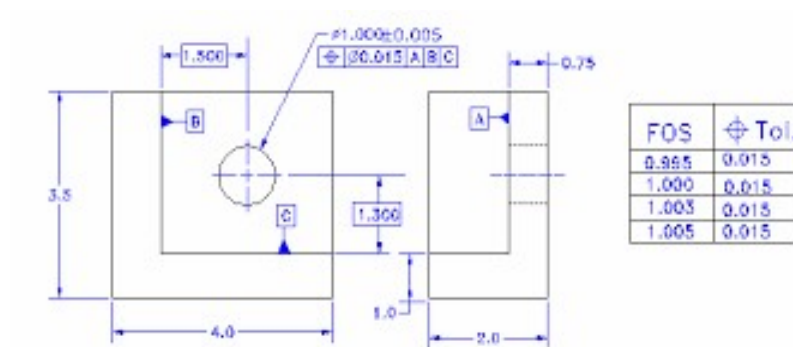
- Circular features with LMC circularity tolerances must conform to the specified form requirements when the feature is at its least material condition, regardless of its actual size.

4. Cylindricity (Cyl):

- Cylindricity with LMC ensures that the specified tolerance for the overall form of the cylindrical feature applies when the feature is at its least material condition.

The LMC concept is often used when the function of the part requires a certain level of form control, especially in situations where variations in size might affect the part's performance. It is important to note that LMC is just one modifier used in GD&T, and the choice of modifier depends on the design intent and functional requirements of the part.

RFS Concept for orientation tolerances:



In Geometric Dimensioning and Tolerancing (GD&T), the concept of RFS, or "Regardless of Feature Size," can be applied to orientation tolerances. When the symbol "RFS" is specified in

the feature control frame for an orientation tolerance, it means that the specified orientation requirement applies without regard to the size of the feature.

For orientation tolerances, such as angularity or parallelism, the RFS modifier ensures that the specified tolerance is independent of the actual size of the feature. This is important because it allows for consistent interpretation of orientation tolerances across different-sized features.

Here's how the RFS concept works for some common orientation tolerances:

1. **Angularity (α):**

- If angularity is specified with the RFS modifier, it means that the angular tolerance applies uniformly regardless of the size of the feature. The angle requirement remains constant irrespective of variations in feature size.

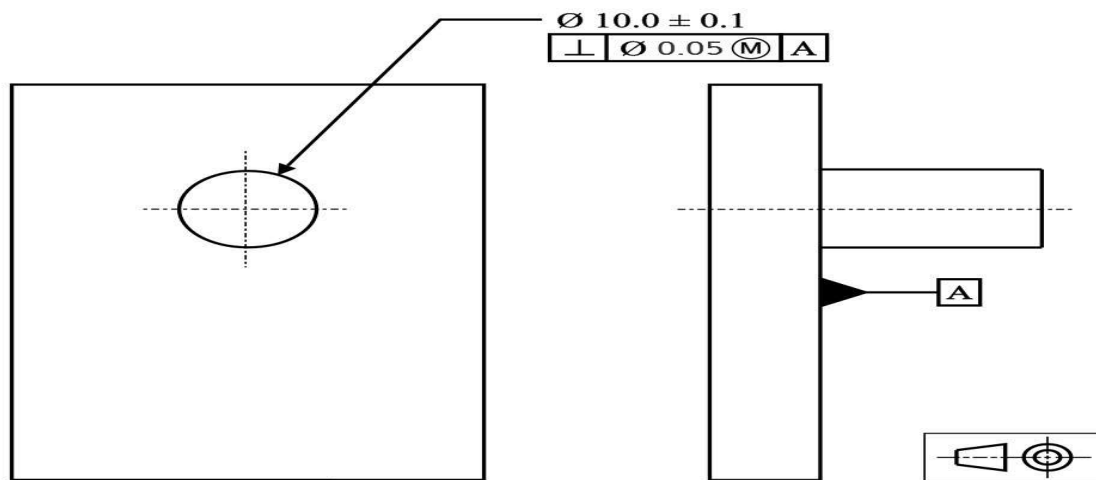
2. **Parallelism ($\parallel$):**

- Parallelism with the RFS modifier ensures that the specified tolerance for the alignment of features remains consistent regardless of the size of the features being compared.

The RFS concept simplifies the application of orientation tolerances and eliminates the need to adjust tolerances based on the size of the part. This can be particularly useful in situations where the functional requirements related to orientation are not influenced by the size of the features.

It's important to note that the choice between RFS and other modifiers, such as MMC (Maximum Material Condition) or LMC (Least Material Condition), depends on the design intent and functional requirements of the part. Each modifier provides a different perspective on how the tolerance should be interpreted in relation to the size of the feature. The selection of the appropriate modifier should be made based on a thorough understanding of the part's function and the desired level of control over orientation.

MMC Concept for orientation Tolerances:



In Geometric Dimensioning and Tolerancing (GD&T), Maximum Material Condition (MMC) can be applied to orientation tolerances in a feature control frame. When MMC is specified for an orientation tolerance, it means that the specified orientation requirement applies when the feature is in its maximum material condition. This implies that the orientation tolerance is based on the assumption that the feature contains the maximum amount of material permissible within the specified limits.

For orientation tolerances, such as angularity or parallelism, specifying MMC in the feature control frame indicates that the tolerance is only applicable when the feature is at its largest size (maximum material condition). This concept is often used to ensure that parts with varying sizes meet specific orientation requirements, especially when the functional aspects of the part are influenced by the orientation of features.

Here's how MMC works for some common orientation tolerances:

1. Angularity (α):

- If angularity is specified with MMC, it means that the angular tolerance applies when the feature is at its maximum material condition. The angle requirement is based on the assumption that the feature contains the maximum amount of material.

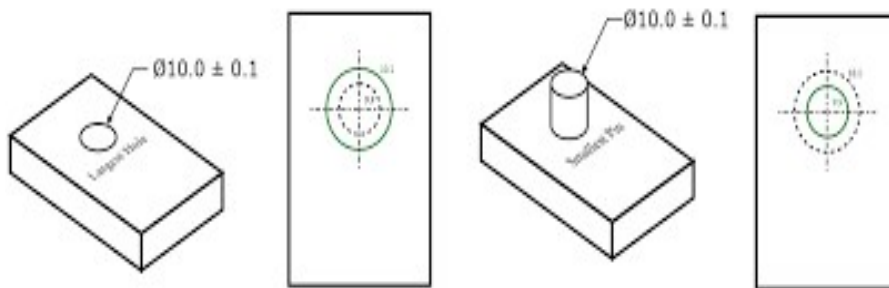
2. Parallelism ($\parallel$):

- Parallelism with MMC ensures that the specified tolerance for the alignment of features applies when the feature is at its maximum material condition.

The MMC concept is useful when the function of the part requires a certain level of orientation control, especially in situations where variations in size might affect the part's performance. It allows for the orientation tolerance to be adjusted based on the assumption that the feature is at its largest size.

It's essential to note that the choice between MMC and other modifiers, such as RFS (Regardless of Feature Size) or LMC (Least Material Condition), depends on the design intent and functional requirements of the part. Each modifier provides a different perspective on how the tolerance should be interpreted in relation to the size of the feature. The appropriate modifier should be selected based on a thorough understanding of the part's function and the desired level of control over orientation.

LMC Concept for orientation tolerances:



In Geometric Dimensioning and Tolerancing (GD&T), LMC stands for Least Material Condition. When LMC is applied to orientation tolerances in a feature control frame, it means that the specified orientation requirement applies when the feature is in its least material condition. This implies that the orientation tolerance is based on the assumption that the feature contains the least amount of material permissible within the specified limits.

For orientation tolerances, such as angularity or parallelism, specifying LMC in the feature control frame indicates that the tolerance is only applicable when the feature is at its smallest size (least material condition). This concept is often used to ensure that parts with varying sizes meet specific orientation requirements, especially when the functional aspects of the part are influenced by the orientation of features.

Here's how LMC works for some common orientation tolerances:

1. **Angularity (α):**

- If angularity is specified with LMC, it means that the angular tolerance applies when the feature is at its least material condition. The angle requirement is based on the assumption that the feature contains the least amount of material.

2. **Parallelism ($\parallel$):**

- Parallelism with LMC ensures that the specified tolerance for the alignment of features applies when the feature is at its least material condition.

The LMC concept is useful when the function of the part requires a certain level of orientation control, especially in situations where variations in size might affect the part's performance. It allows for the orientation tolerance to be adjusted based on the assumption that the feature is at its smallest size.

It's essential to note that the choice between LMC and other modifiers, such as RFS (Regardless of Feature Size) or MMC (Maximum Material Condition), depends on the design intent and functional requirements of the part. Each modifier provides a different perspective on how the tolerance should be interpreted in relation to the size of the feature. The appropriate modifier should be selected based on a thorough understanding of the part's function and the desired level of control over orientation.

Interpretation of Form And orientation tolerances in GD&T

Squareness:

In this example, there are two squareness examples. Figure 1 and figure 2 show the example of the squareness of an axis (line) with regards to a datum axis and the example of the squareness of a surface with respect to a datum axis, respectively.

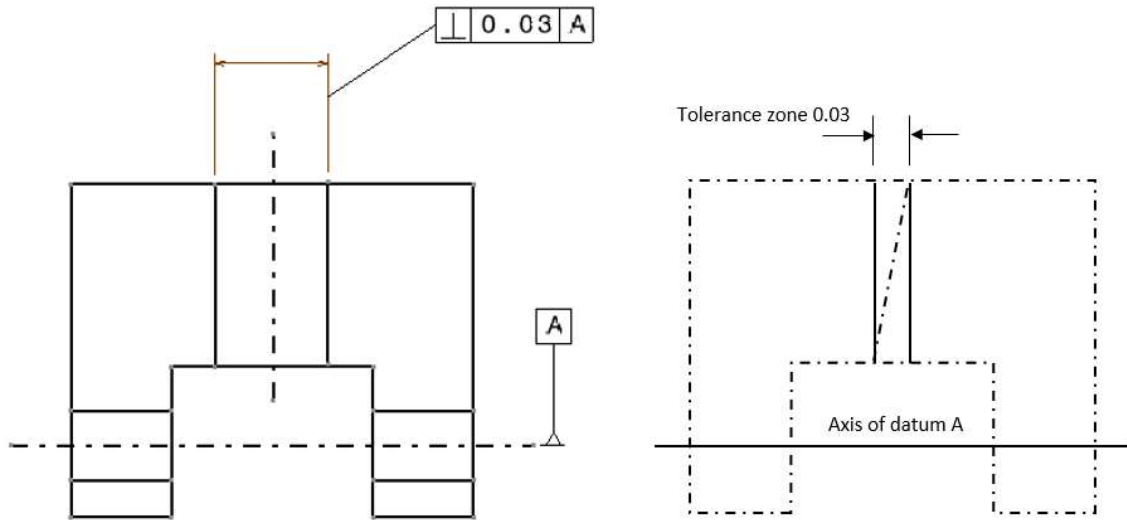


Figure 1: Example of the squareness of an axis line with regards to the axis of a datum. (left) the technical drawing and (right) the GD&T interpretation.

The reading and interpretation of the example in figure 1 above is as follow. The axis of the pointed vertical hole should lie within its tolerance zone. This tolerance zone is two hypothetical parallel planes that are separated for 0.03 (unit distance) and these parallel planes are perpendicular with respect to the axis of datum A.

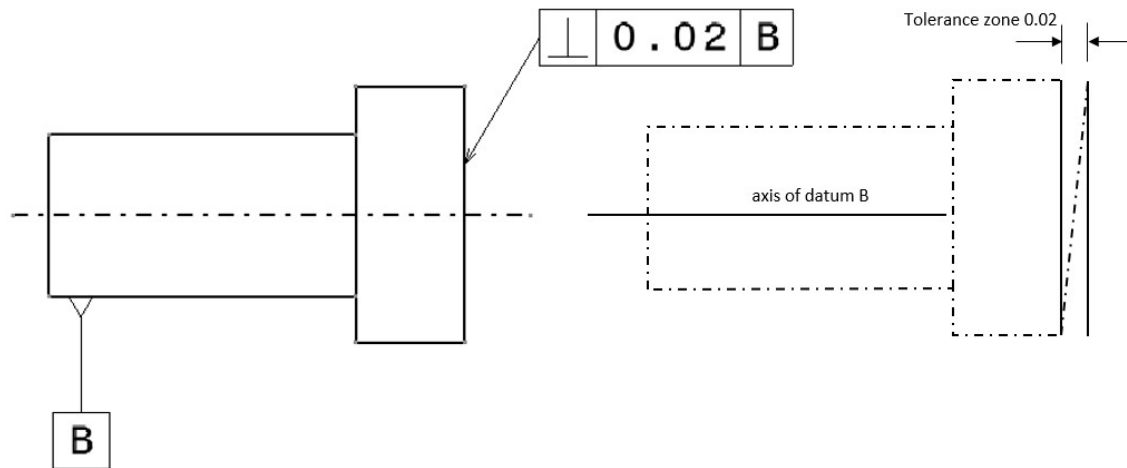


Figure 2: Example of the squareness of a surface with regards to the axis of a datum. (left) the technical drawing and (right) the GD&T interpretation.

The reading and interpretation of the example in figure 2 above is as follow. The pointed cylinder surface should lie within its tolerance zone. This tolerance zone is defined as two

hypothetical parallel planes that are separated for 0.02 (unit distance) and are perpendicular with respect to the axis of datum B.

Parallelism:

Examples of parallelism are shown in figure 3 and figure 4. In figure 3 below, the example of the parallelism of a surface with respect to a datum surface is shown. The interpretation of the parallelism tolerance in figure 3 is as follow.

The pointed surface should lie within a tolerance zone that is two parallel planes that are separated for 0.02 and are parallel with respect to the surface of datum A.

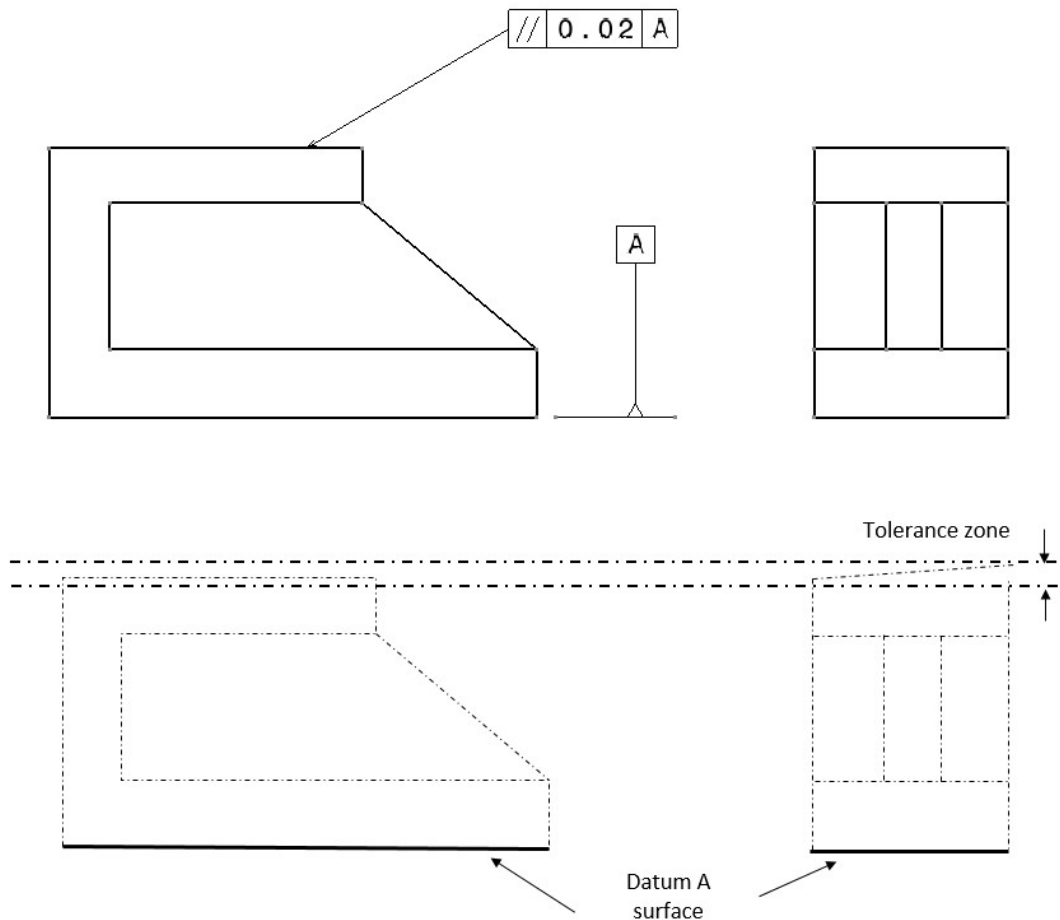


Figure 3: Example of parallelism of a surface with respect to a datum surface. (top) the technical drawing of the part and (bottom) the interpretation of the parallelism tolerance.

In figure 4 below, the example of the parallelism of an axis with respect to a datum axis is presented. The parallelism interpretation of figure 4 is as follow.

The axis of the pointed hole should lie within a tolerance zone that has a cylindrical shape with diameter of 0.03 (unit distance). The cylindrical tolerance zone should be perpendicular with respect to the axis of datum A.

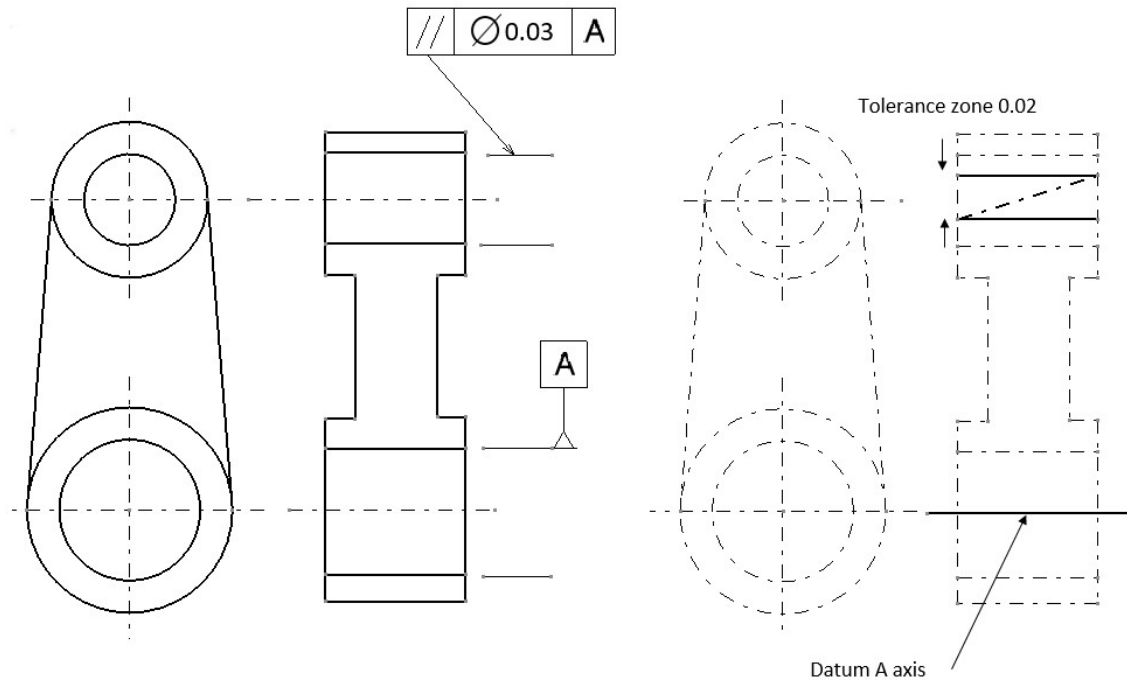


Figure 4: Example of the parallelism of an axis with respect to a datum axis. (left) the technical drawing of the part and (right) the interpretation of the parallelism tolerance.

From the two examples of parallelism tolerance, there are various types of parallelism. For example, a parallelism between an axis and a surface, between a surface with another surface and between one axis with another axis. These various types of situations are also found in other types of GD&T tolerance.

Line profile:

Figure 5 below shows the example of the line profile tolerance of a curve on a part. The geometrical interpretation from the technical drawing in figure 5 is as follow.

The pointed line profile on the drawing should lie within a tolerance zone. This tolerance zone is two line profiles that are equally separated on each point in the line for 0.1 (unit distance).

Line profile is **a non-related tolerance**. That is, this type of tolerance does not need a datum definition (similar to flatness, roundness or circularity and cylindricity tolerance).

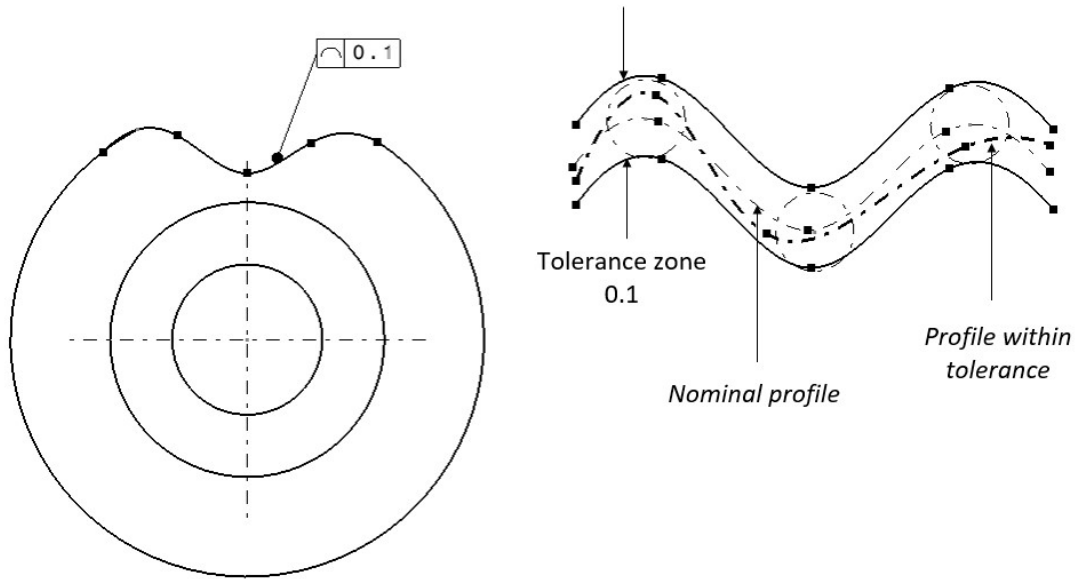


Figure 5: Example of the profile tolerance of a curve of a freeform surface. (left) the technical drawing of the part's profile and (right) the interpretation of the profile tolerance.

Surface profile:

The example of a surface profile tolerancing on a part with curve-surface is presented in figure 6. In figure 6, the reading and interpretation of the surface profile tolerance is as follow. The pointed surface on the part should be within a tolerance zone. This tolerance zone is defined as two curved surfaces (identical to the nominal curve surface of the part) that are separated equally at any point on the surface (parallel) with a distance of 0.02 (unit distance).

Since this tolerance is a type of form tolerance (such as flatness, straightness, roundness or circularity and cylindricity), this surface profile tolerance does not require any datum (un-related tolerance).

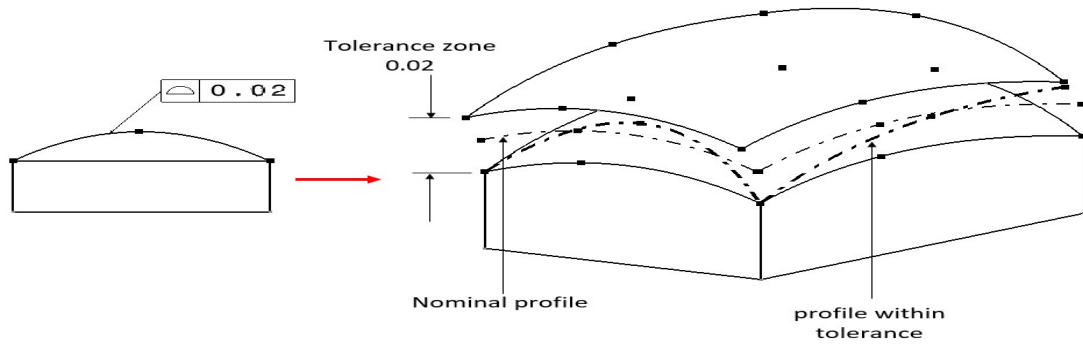


Figure 6: Example of the surface profile tolerance with a curve shape on a component. (left) the technical drawing with the surface profile tolerance symbol and (right) the interpretation of the surface tolerance symbol on the part.

One important thing to know is that, because the shape of the surface is curved, the measurement process (which commonly performed by using a coordinate measuring machine/CMM) to measure the curved surface and verify the tolerance is significantly affected by how many measurement points taken from the surface. The sampling strategy of the measurement should be good enough to be able to accurately reconstruct the curved surface and extract the measurement result.

Symmetry:

There are two symmetry tolerance examples that are shown in figure 7 and figure 8. In figure 7, the example of symmetry tolerance of an axis with respect to a datum axis is presented.

The symmetry tolerance interpretation in figure 7 below is as follow. The pointed axis should be within two parallel lines that are separated each other for 0.08 (unit distance). These parallel lines should contain the averaged-axis line from datum A and datum B. This averaged-line should be at the middle between the two parallel lines.

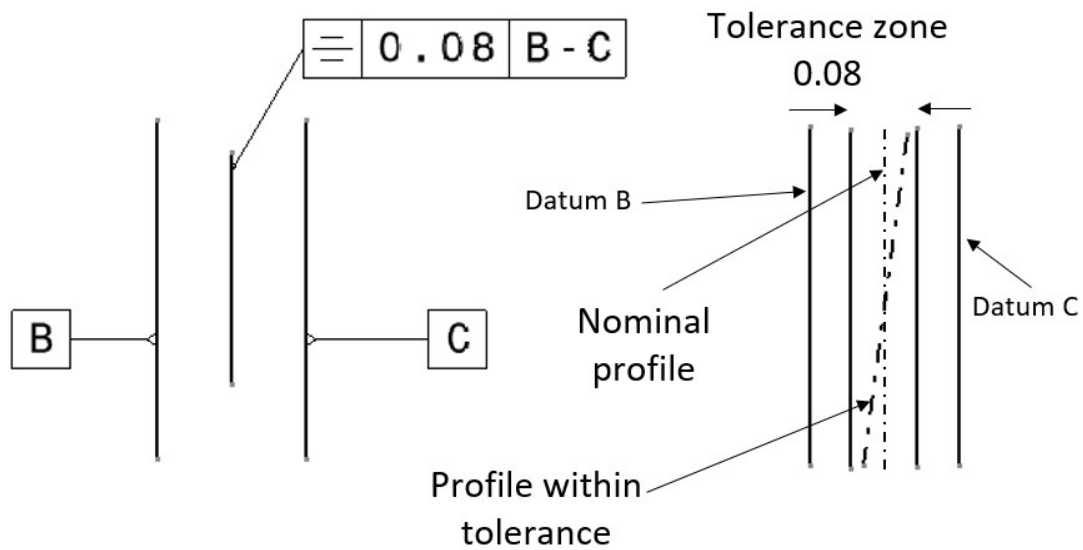


Figure 7: Example of the symmetry tolerance of an axis with respect to a datum axis. (left) the technical drawing and (right) the symmetry symbol interpretation from the drawing.

Figure 8 below presents the example of symmetry tolerance of an axis with respect to an averaged-axis from two datum.

The interpretation of symmetry in figure 8 is as follow. The pointed axis should be within two parallel lines that are separated each other for 0.03 (unit distance). These two parallel lines should contain an averaged-line from the axis of datum A and datum B at the middle.

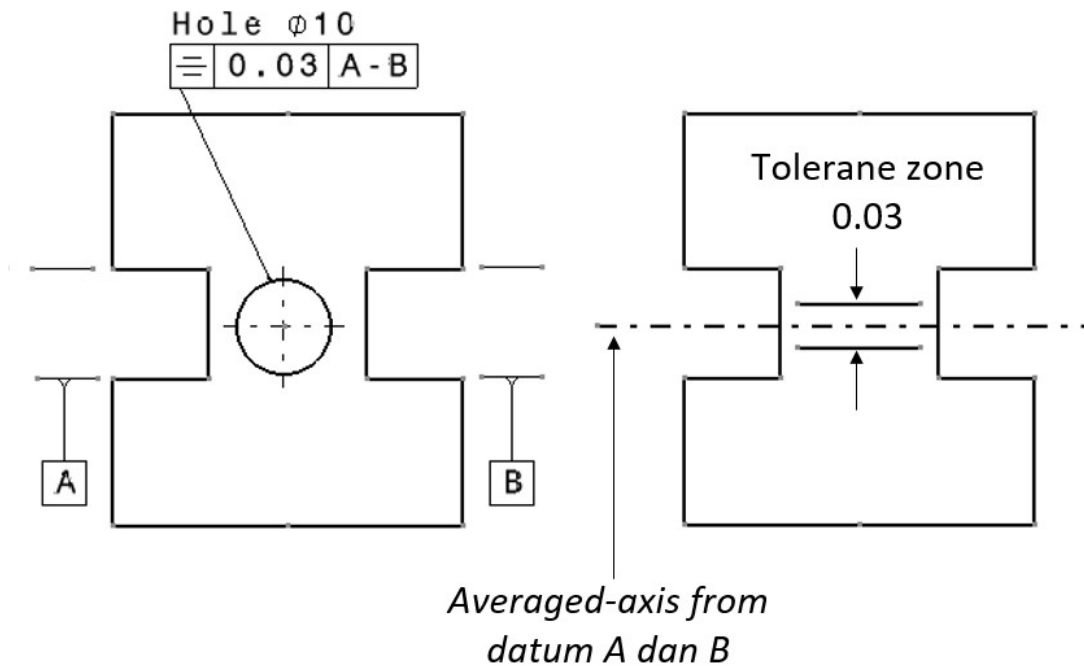


Figure 8: Example of the symmetry tolerance of an axis with respect to an averaged-axis from two datum. (left) the technical drawing of symmetry tolerance and (right) the interpretation of the symmetry symbol from the drawing.

Concentricity:

Figure 9 and 10 below shows the example of concentricity of an axis with respect to a datum axis and with respect to an averaged-axis from two datum A-B, respectively.

The interpretation of concentricity in figure 9 below is as follow. The axis of the pointed cylinder in the technical drawing should be within a tolerance zone in a form of cylinder with diameter of 0.03 (unit distance). The axis of the cylindrical tolerance zone should be co-axial with the axis of datum A cylinder.

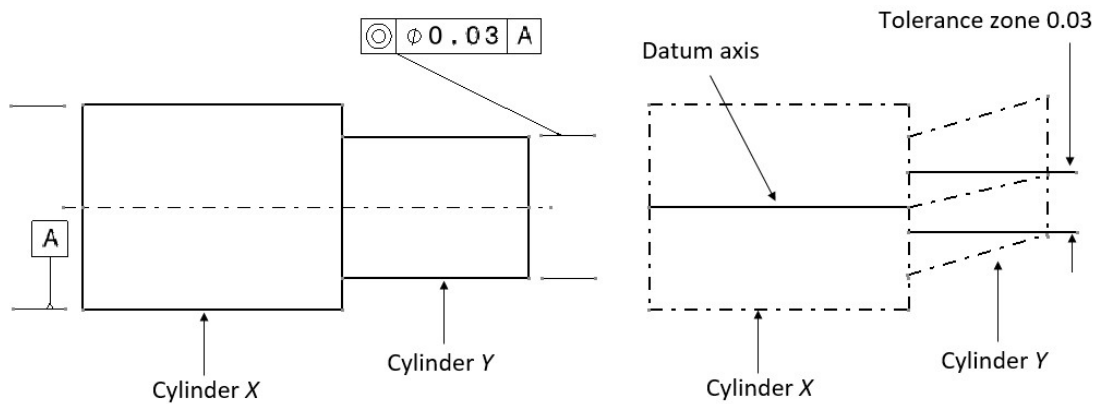


Figure 9: Example of the concentricity tolerance of an axis with respect to a datum axis. (left) the technical drawing and (right) the interpretation of the concentricity symbol.

Meanwhile, the interpretation of concentricity in figure 10 below is as follow. The axis of the pointed cylinder in the drawing (figure 10) should be within a tolerance zone. This tolerance zone should be a cylinder with a diameter of 0.05 (unit distance) and is co-axial with the averaged-axis of datum A cylinder and datum B cylinder.

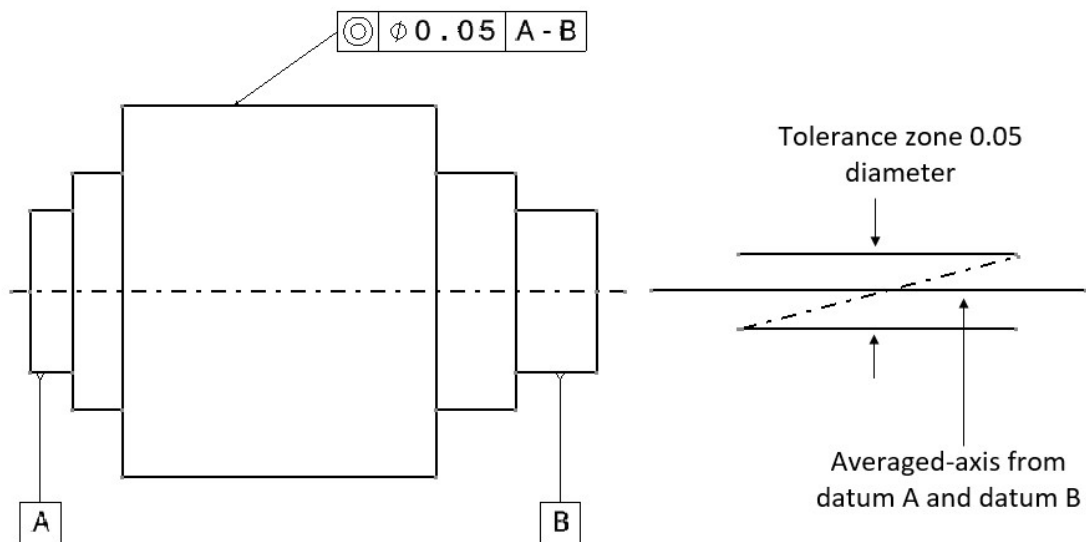


Figure 10: Example of the concentricity tolerance of an axis with respect to an averaged-axis from two datum A-B. (left) the technical drawing and (right) the concentricity symbol interpretation from the drawing.

Angularity:

There are two examples of angularity tolerance shown in figure 11 and figure 12 below. In figure 11, angularity of an axis is presented. Meanwhile in figure 12, angularity of a surface plane is presented.

The interpretation of angularity tolerance in figure 11 is as follow. The axis of the pointed hole in the drawing should be within a cylindrical tolerance zone with a diameter of 0.06 (unit distance). The axis of this cylindrical tolerance zone should has orientation of 60 degree with respect to the surface of datum A.

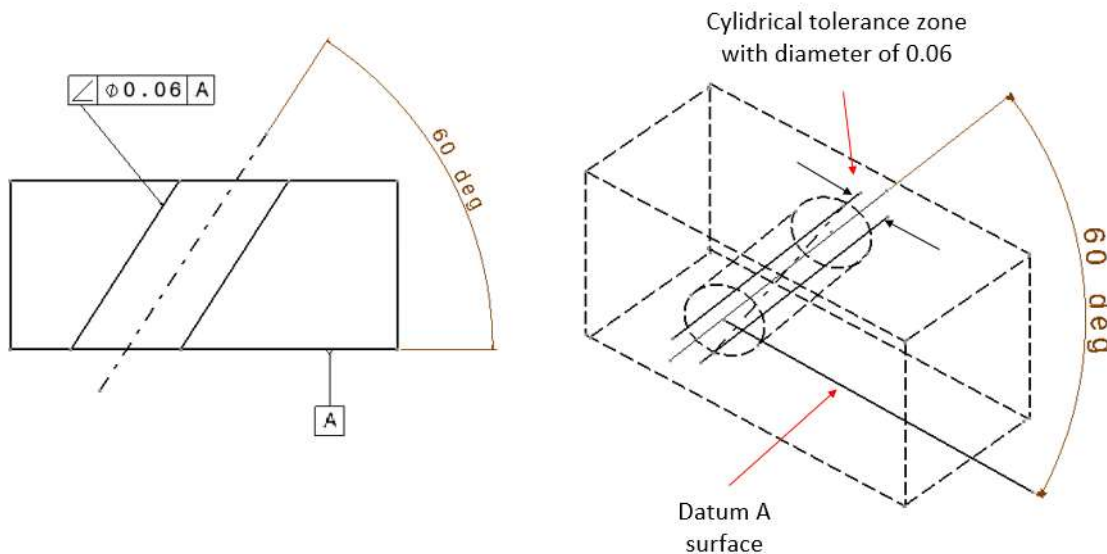


Figure 11: Example of the angularity tolerance of an axis with respect to a surface of datum A. (left) the angularity symbol and (right) the angularity tolerance interpretation.

Meanwhile, the interpretation of angularity in figure 12 below is as follow. The pointed surface on the technical drawing should be within a tolerance zone. This tolerance zone is two parallel planes that are separated for 0.08 (unit distance). These two parallel planes should be oriented for 30 degree with respect to datum A surface shown in the drawing (figure 12).

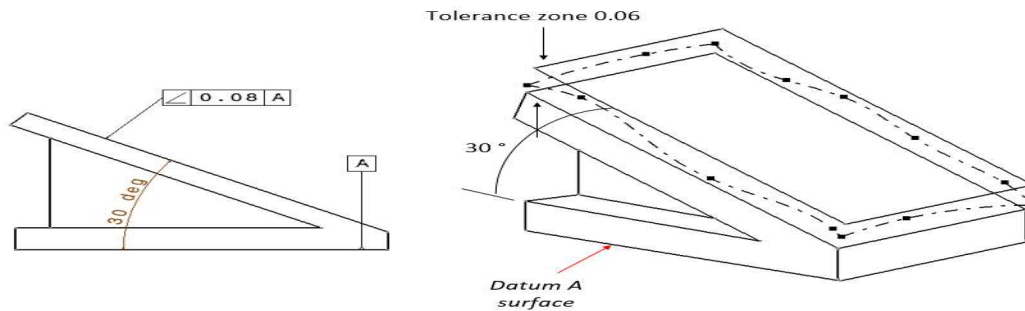


Figure 12: Example of the angularity tolerance of a surface with respect to a datum A surface. (left) the drawing showing the angularity symbol and (right) the interpretation of the angularity symbol.

Position:

Figure 13 and figure 14 below shows the examples of position tolerance of an axis with respect to other two axes and with respect to a cylinder axis, respectively.

The explanation of position tolerance in figure 13 is as follow. The axis of the pointed hole should be within a cylindrical tolerance zone with diameter of 0.01 (unit distance). The axis distance of this cylindrical tolerance zone should be 30 (unit distance) with respect to the first cylinder and 45 (unit distance) with respect to the second cylinder (figure 13 right). In addition, the axis of the cylindrical tolerance zone should be parallel with the axis of the other two cylinders (holes).

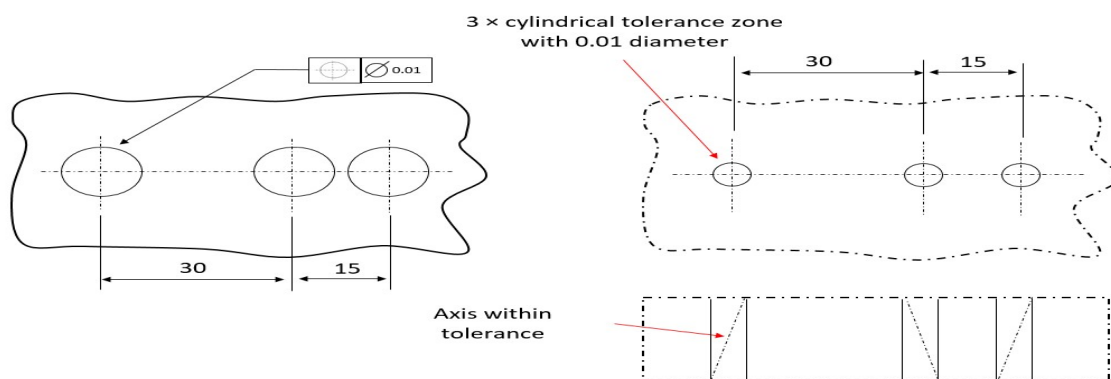


Figure 13: Example of the position tolerance of the axis of a hole with respect to other hole axes. (left) the technical drawing of the position tolerance and (right) the interpretation of the position tolerance from the drawing.

The explanation of position tolerance in figure 14 is as follow. The axis of the pointed hole should be within a tolerance zone. This tolerance zone has a cylindrical shape with diameter of 0.05 (unit distance). Also, the axis of the cylindrical tolerance zone should have a distance of $50/2=25$ (unit distance) from the axis of the main (the biggest) cylinder (figure 14 left).

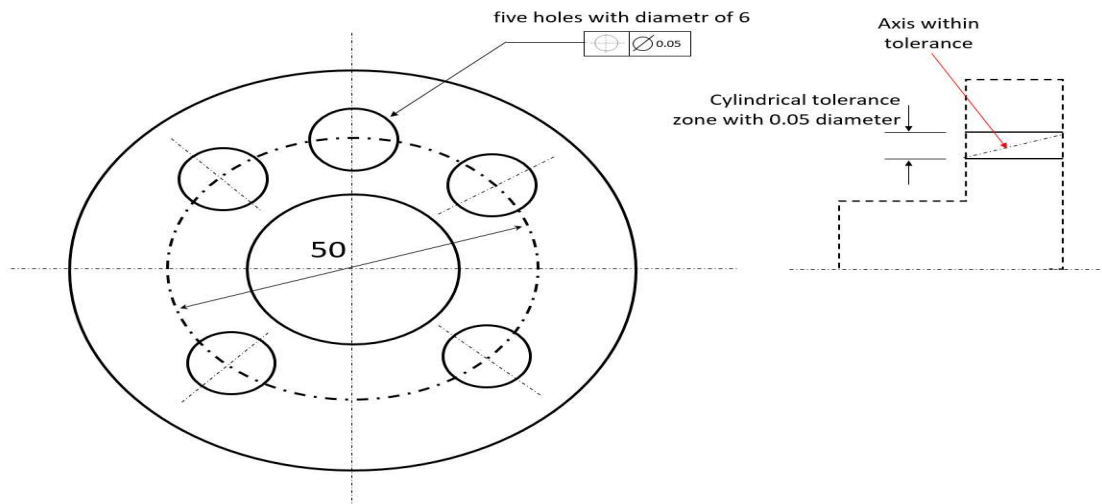


Figure 14: Example of the position tolerance of the axis of a hole with respect to a cylinder axis. (left) the position tolerance in its technical drawing and (right) the position tolerance interpretation from the drawing.

Specifically for position (or location) tolerance, although its symbol in a technical drawing does not refer to a datum, inherently, position/location tolerance symbol refer to a datum. That is why, for clarity, in general, position/location tolerance symbol in a technical drawing also shows a reference datum.

Straightness:

Examples of straightness on the same part, but with different tolerance conditions are presented in figure 15 below.

In figure 15a below, the example of straightness of the pointed cylindrical surface and not the axis of the cylinder. The interpretation of this straightness is as follow. The profile of the pointed surface should be within a tolerance zone of two parallel lines separated for 0.03 unit distance. The measurement for this tolerance is by taking points of profile lines by a CMM or other measuring instruments.

In figure 15b below, the example shows the straightness of the pointed axis of the small cylinder. The reading of this straightness tolerance is as follow. The axis of the pointed small cylinder should be within a tolerance zone that is two parallel lines separated for 0.03. In addition, the length of this tolerance zone follows the length of the small cylinder only (not including the big cylinder).

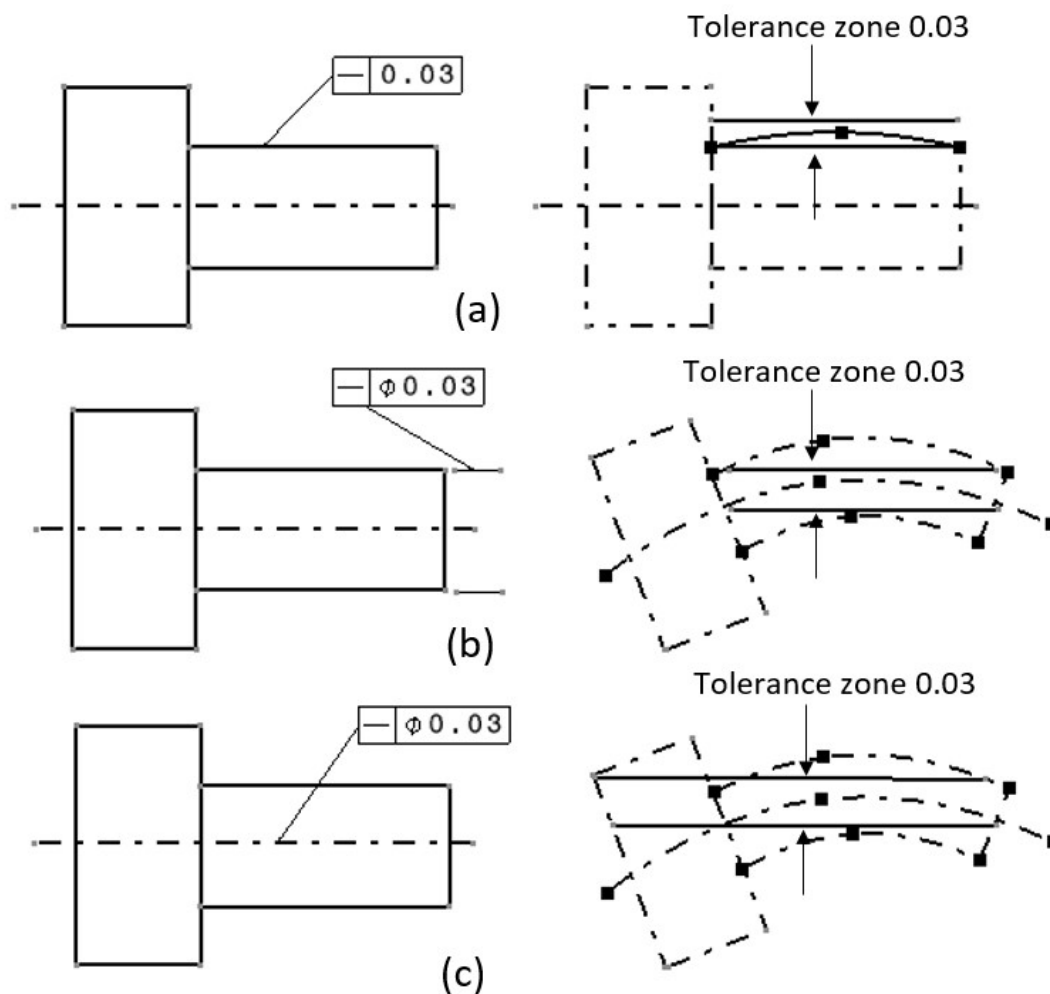


Figure 15: (a) Example of the straightness of a cylindrical surface, (b) Example of the straightness of the axis of a cylinder attached to another cylinder and (c) Example of the straightness of the axis of two cylinders with different diameter as a whole.

In figure 15c above, the example shows the straightness of an axis with respect to the whole length of the cylinder, including both the small and big cylinders. The interpretation of the straightness tolerance in figure 15c above is as follows. The whole axis of the cylinder (both the small and the big one) should be within a tolerance zone. This tolerance zone is in the form

of two parallel lines that are separated for 0.03 unit distance and the length of this tolerance zone (the two parallel lines) covers the whole length of the cylinder (both the small and the big one).